Adding Type Annotations to Classes

TypeScript takes full advantage of the class features added to JavaScript in ES6. (There are a lot of references for ES6 classes; the basic syntax is available online.)[50] The goal of the TypeScript extensions to class syntax are to allow the TypeScript compiler to treat object attribute references and method calls the way functions and assignments are treated. We want to be able to tell from the class of the instance what attributes exist and the expected type of those attributes.

In TypeScript, any method defined in a class is available to instances of the class. You can annotate arguments and return values of a method just as you would for functions.

The first real change in TypeScript classes compared to JavaScript is the need to explicitly list attributes of the class that would, in plain JavaScript, only be created when assigned. In TypeScript, if you are going to refer to an attribute like `this.color = "red"`, the existence of the `color` attribute needs to have already been declared.

We've already seen this in our Stimulus controllers, where we explicitly need to specify the properties that Stimulus is going to create so that the TypeScript will let us use them:

chapter_05/07/app/javascript/controllers/css_controller.ts

```
import { Controller } from "@hotwired/stimulus"

export default class CssController extends Controller {
  static classes = ["css"]
  cssClasses: string[]

  static targets = ["elementToChange"]
  elementToChangeTarget: HTMLElement

  static values = { status: Boolean }
  statusValue: boolean

  toggle(): void {
```